



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

January 8, 2025

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page must be filled with your **name, surname, and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam, you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets, and/or similar. As well, you are not allowed to bring with you any notes, books, written schemes, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You can write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room no earlier than half the time of the duration of the exam. You are not allowed to keep the exam text with you while leaving the room.
- **Three of the points will be given based on how quick you are in solving the exam**. If you finish earlier than 30 min before the end of the exam, you will get 3 points; if you finish earlier than 20 min, you will get 2 points; and if you finish earlier than 10 min, you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Derive the **Bias-Variance Decomposition** for regression under the squared error loss, considering a target variable y , a prediction model $\hat{f}(x)$, and the true function $f(x)$. Explain how bias and variance contribute to the overall expected error of the model.

Exercise 2 (7 marks)

What is the **VC dimension** of a hypothesis space? What can it be used for?

Exercise 3 (3 marks)

Consider the following snippet of code:

```
1 X1 = zscore(dataset[['sepal-length', 'sepal-width']].values)
2 X2 = zscore(dataset[['sepal-length', 'sepal-width', 'petal-length']].values)
3 t = dataset['class'].values == 'Iris-setosa'
4
5 model1 = LogisticRegression(penalty='none')
6 model1.fit(X1, t)
7 accuracy1 = model1.score(X1, t)
8
9 model2 = LogisticRegression(penalty='none')
10 model2.fit(X2, t)
11 accuracy2 = model2.score(X2, t)
12
13 best_model = model1 if accuracy1 > accuracy2 else model2
14
```

1. Describe the procedure performed in the code above, detailing **line-by-line** the operations and their purpose.
2. Is the process used in the code correct? Explain why or why not.
3. Can Ordinary Least Squares (OLS) be applied to the same problem? Discuss whether OLS is suitable for classification tasks and explain its limitations compared to logistic regression.

1. The code implements two logistic regression models, one using two input variables (**sepal-length** and **sepal-width**) and the other using three (**sepal-length**, **sepal-width**, and **petal-length**). Both input matrices are normalized using **zscore** (lines 1 and 2). The target variable (**t**) represents a binary classification problem for identifying **Iris-setosa**. Each model is trained on its respective dataset, and training accuracy is computed (lines 5–8, 9–11). The model with the higher accuracy is selected as the best model (line 13).
2. The process of using training set accuracy to select the best model is flawed. It does not generalize well to unseen data, as the accuracy on the training set often overestimates the true performance of the model. Instead, cross-validation or a separate validation set should be used to select the best model.
3. OLS can technically be applied to the same problem by interpreting the binary classification as a regression problem where the target values are 0 and 1. However, OLS is not a good idea for classification, as it does not model the probability distribution of the target variable correctly. Logistic regression is specifically designed for binary classification and provides probabilistic outputs, making it a better choice.

Exercise 4 (2 marks)

Determine whether the following statements about the Principal Component Analysis (PCA) procedure are true or false. Motivate your answers.

1. The set of the Principal Components vectors forms an orthonormal basis for the original feature space.
2. Projecting the original features onto the Principal Components and using these projections as features for regression or classification reduces the likelihood of overfitting.
3. The percentage of the variance explained by a Principal Component is inversely proportional to the value of the corresponding eigenvalue.
4. The procedure to apply PCA to a dataset is deterministic.

1. TRUE: The procedure for the PCA looks for the direction in the dataset which is providing the most variance and extracts the (first) Principal Component (PC) as the unit vector identifying that direction. Iteratively, it checks the direction, orthogonal to the previous one, with the most variance and extracts another (second) PC. The process iterates over a number of PC equal to the number of dimensions of the dataset. This produces an orthogonal base to the initial dataset.
2. FALSE: only selecting the first K PC one has the chance to remove the noise from the data and avoid overfitting. If we are keeping all the PC, we would have a linear transformation of the original dataset, which is likely to behave as good as the original one for the supervised task.
3. FALSE: The variance explained by each PC is directly proportional to the corresponding eigenvalue. Indeed, its formula is $\frac{\lambda_i}{\sum_i \lambda_i}$, where λ_i is the eigenvalues corresponding to the i -th PC.
4. TRUE: The procedure includes an input normalization procedure and computing the eigenvectors of the covariance matrix, which are both deterministic procedures.

Exercise 5 (2 marks)

Determine whether the following statements about the Multi-Armed Bandit (MAB) problem are true or false. Provide adequate motivations.

1. In a stochastic MAB setting, the reward distribution of each arm remains fixed throughout the learning procedure.
2. MAB techniques can be directly applied to problems where there are dependencies between arms.
3. The exploration-exploitation trade-off is a fundamental challenge in MAB problems.
4. The Thompson Sampling (TS) algorithm requires the knowledge of the reward distributions of the arms in advance.

1. TRUE: In the stochastic MAB setting, the reward distribution of each arm does not change over time, and the learner's task is to learn these distributions.
2. FALSE: Standard MAB techniques assume independent arms. Dependencies between arms require specialized algorithms, such as those for structured bandit problems.
3. TRUE: The core challenge in MAB is balancing exploration (to discover better arms) and exploitation (to maximize reward based on current knowledge).
4. FALSE: Thompson Sampling does not require prior knowledge of reward distributions but instead uses a prior belief updated with observed rewards to sample actions.

Exercise 6 (2 marks)

You are an ML specialist working for an agricultural robotics company that develops drones to monitor and spray crops in a large 2D field. Consider the following problems:

1. Predict the time required for a drone to reach a specific area in the field based on its current position and wind speed using a dataset of past drone flights.
2. Learn a drone navigation policy that minimizes the time and fuel consumption needed to spray all the crops in the field, assuming you can simulate the drone's operation.

Model the two problems described above as ML problems, specifying the data (features and target or states, actions, and reward), the problem class, and a specific method to solve them.

1. **SUPERVISED LEARNING - REGRESSION.** The features are the current position of the drone (x, y) , wind speed and direction, and the position of the target area. The target is the time needed to reach the target area. A method to solve it is Support Vector Regression (SVR).
2. **REINFORCEMENT LEARNING - CONTROL.** The state consists of the drone's current position, remaining fuel, and the positions of unsprayed crops. The actions are the direction and velocity of movement. The reward could be a negative constant for each time step, plus a larger penalty for running out of fuel. The problem is episodic, ending when all crops are sprayed. A possible method to solve this problem is Q-Learning or SARSA.

Exercise 7 (4 marks)

Consider the following trajectory obtained by an agent interacting with an MDP having three states $\mathcal{S} = \{X, Y, Z\}$ and two actions $\mathcal{A} = \{\text{yes}, \text{no}\}$,

$(X, \text{yes}, 3) \rightarrow (Y, \text{no}, -1) \rightarrow (Z, \text{yes}, 2) \rightarrow (X, \text{no}, 4) \rightarrow (Y, \text{yes}, 0) \rightarrow (Z, \text{no}, 1) \rightarrow (Y, \text{no}, -2) \rightarrow (X, \text{yes}, 5)$.

Consider initial state-action values $Q(s, a) = 0$ for every state-action pair, a learning rate $\alpha = 0.2$, and a discount factor $\gamma = 1$. Answer the following questions, providing adequate motivations.

1. Execute the *Q-learning* algorithm on the given trajectory (break ties using the first action listed in the action set).
2. Execute the *SARSA* algorithm on the given trajectory.
3. Derive the best policy for each state according to the output of *SARSA*. Is it different from the one provided by *Q-learning*?

1. The formula for Q-learning is $Q(s, a) \leftarrow Q(s, a) + \alpha(R + \gamma \max_{a'} Q(s', a') - Q(s, a))$. Applying it step by step:

$$Q(X, \text{yes}) = 0 + 0.2(3 + 1 \cdot 0 - 0) = 0.6$$

$$Q(Y, \text{no}) = 0 + 0.2(-1 + 1 \cdot 0 - 0) = -0.2$$

$$Q(Z, \text{yes}) = 0 + 0.2(2 + 1 \cdot 0.6 - 0) = 0.52$$

$$Q(X, \text{no}) = 0 + 0.2(4 + 1 \cdot 0 - 0) = 0.8$$

$$Q(Y, \text{yes}) = 0 + 0.2(0 + 1 \cdot 0.52 - 0) = 0.104$$

$$Q(Z, \text{no}) = 0 + 0.2(1 + 1 \cdot 0.104 - 0) = 0.2208$$

$$Q(Y, \text{no}) = -0.2 + 0.2(-2 + 1 \cdot 0.8 - (-0.2)) = -0.4$$

2. The formula for SARSA is $Q(s, a) \leftarrow Q(s, a) + \alpha(R + \gamma Q(s', a') - Q(s, a))$, where a' is the action taken in the next step. Applying it step by step:

$$Q(X, \text{yes}) = 0 + 0.2(3 + 1 \cdot 0 - 0) = 0.6$$

$$Q(Y, \text{no}) = 0 + 0.2(-1 + 1 \cdot 0 - 0) = -0.2$$

$$Q(Z, \text{yes}) = 0 + 0.2(2 + 1 \cdot 0 - 0) = 0.4$$

$$Q(X, \text{no}) = 0 + 0.2(4 + 1 \cdot 0 - 0) = 0.8$$

$$Q(Y, \text{yes}) = 0 + 0.2(0 + 1 \cdot 0 - 0) = 0$$

$$Q(Z, \text{no}) = 0 + 0.2(1 + 1 \cdot (-0.2) - 0) = 0.16$$

$$Q(Y, \text{no}) = -0.2 + 0.2(-2 + 1 \cdot 0.6 - (-0.2)) = -0.44$$

3. The policy prescribed by SARSA is: $X \rightarrow \text{no}$, $Y \rightarrow \text{yes}$, $Z \rightarrow \text{yes}$. The policy from Q-learning is: $X \rightarrow \text{no}$, $Y \rightarrow \text{yes}$, $Z \rightarrow \text{yes}$. Therefore, the policies differ.

Exercise 8 (4 marks)

Consider a linear binary SVM classifier that, after training, has weights $w = [2, -1]$ and bias $b = 1$. Motivate all answers and write down any necessary computation.

1. How is the point $[2, 3]$ classified?
2. Provide an example of a support vector for this SVM.
3. Suppose the point $[3, 1]$ is added to the dataset with a positive label. Do we need to retrain the SVM?
4. Can we discard the point $[0, -2]$ from the dataset if all we care about is predicting the class of new points?

1. $f([2, 3]) = 2 \cdot 2 - 1 \cdot 3 + 1 = 2 > 0$, so the point is classified as positive.
2. An example is $[0, 1]$, since $f([0, 1]) = 2 \cdot 0 - 1 \cdot 1 + 1 = 0$, so it is **within** the boundary. Any vector v with $|f(v)| \leq 1$ is a good example.
3. No, since $f([3, 1]) = 2 \cdot 3 - 1 \cdot 1 + 1 = 6 > 1$: the point is correctly classified and is not a support vector, so the classifier would not change.
4. With a linear SVM, we can discard the whole dataset; the weights are enough! If we are using a non-parametric version (e.g., a kernel-based implementation with a linear kernel), we have to check if the point is a support vector. $f([0, -2]) = 2 \cdot 0 - 1 \cdot (-2) + 1 = 3 > 1$, so it is not a support vector, and it can be discarded anyway.

